

1. a) $\vec{x} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} + s \underbrace{\begin{pmatrix} 4 \\ -1 \\ -4 \end{pmatrix}}_{\vec{AB}} + t \underbrace{\begin{pmatrix} 2 \\ 2 \\ -8 \end{pmatrix}}_{\vec{AC}}$

b) $AB = 6$; $AC = \sqrt{72}$; $\vec{BC} = \begin{pmatrix} -1 \\ 4 \\ -4 \end{pmatrix}$ $BC = 6$
gleichschwellig ; $6^2 + 6^2 \stackrel{!}{=} 72$, also $\triangle$ bei B

d) $\vec{r}_D = \vec{r}_C - \vec{AB} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ D(0|0|0)

2. $f_a(x) = a \cdot x \cdot (x-5)^2$; $f'_a(x) = a(3x^2 - 10x + 15)$; $f''_a(x) = a(6x - 10)$
 $= a(x^3 - 10x^2 + 15x)$

a) $f'(3) = -\frac{1}{2}$

$a \cdot (3 \cdot 3^2 - 10 \cdot 3 + 15) = -\frac{1}{2} \rightarrow a = \frac{1}{16}$

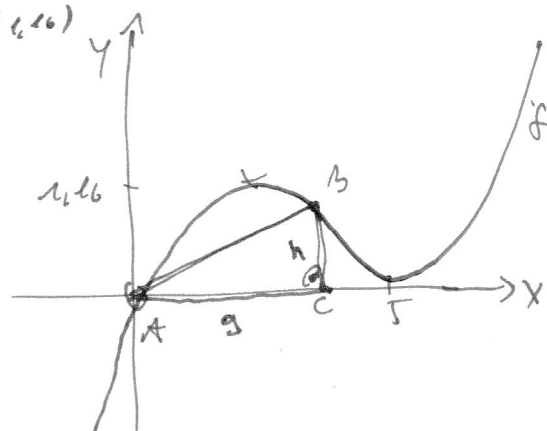
$f(x) = \frac{1}{16} x(x-5)^2$ $f'(x) = \frac{1}{16} (3x^2 - 10x + 15)$; $f''(x) = \frac{1}{8} (6x - 10)$

WS : $x=0$ $x=5$

$f' = 0$
 $x=5$ $y=0$
 $x=5/3$ $y=1/16$

$f'' = 0$
 $x=10/3$ $y=0,58$
eintr. u. w. w. p. $(\frac{10}{3} | 0,58)$

$f''(5) > 0 \Rightarrow \text{Min}(5|0)$
 $f''(5/3) < 0 \Rightarrow \text{Max}(5/3 | 1/16)$



b) $a \cdot x \cdot (x-5)^2 = 0$
WS : 0 5

$A = \int_0^5 f_a(x) dx = a \left[\frac{1}{4} x^4 - \frac{10}{3} x^3 + \frac{15}{2} x^2 \right]_0^5$
 $= a \cdot \frac{625}{12}$

c) $A_{ABC} = \frac{1}{2} gh = \frac{1}{2} x \cdot f_a(x) = \frac{1}{2} a \cdot (x^4 - 10x^3 + 15x^2) \rightarrow \text{max}$

$D = [0; 5]$

$A' = \frac{1}{2} a (4x^3 - 30x^2 + 30x) = 0$
 $x=0$; $x=\frac{5}{2}$; $x=5$

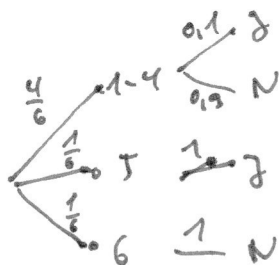
$A''(x) = \frac{1}{2} a (12x^2 - 60x + 30)$

$A''(\frac{5}{2}) < 0 \Rightarrow \text{Max}$ $\left. \begin{matrix} A(0)=0 \\ A(5)=0 \end{matrix} \right\} \text{Min}$ $\text{Max } x = \frac{5}{2}$
 $B(\frac{5}{2} | \frac{125}{8} a)$

3. 1-4: ehrlich; 5: ja; 6: kein $p=0,1$

Passwort

a)



$$P(A \mid j) = \frac{4}{6} \cdot 0,1 + \frac{1}{6} = \frac{7}{30}$$

~~$P(A \mid j)$~~

$$P(B: \text{schummelt} \mid j) = \frac{4/6 \cdot 0,1}{7/30} = \frac{2}{7}$$

$$P(C: \text{wird. 1 v. w.} \mid j)$$

$$= 1 - P(\text{Kein v. w.})$$

$$= 1 - \left(\frac{23}{30}\right)^{10} = \underline{0,93}$$

b) $P(j) = \frac{4}{6} \cdot p + \frac{1}{6}$

c) $P(j) = 22,8\% \approx \frac{4}{6} p + \frac{1}{6}$
 $p = 9\%$

4.1 a) $\begin{pmatrix} -2 \\ 1 \end{pmatrix} = c \cdot \begin{pmatrix} 3 \\ 2h \end{pmatrix} \Rightarrow \begin{matrix} -2 = 3c \Rightarrow c = -\frac{2}{3} \\ 1 = 2c \cdot h \Rightarrow h = \frac{1}{2c} = -\frac{3}{4} \end{matrix}$

b) $\begin{pmatrix} 1 \\ 4 \\ u-1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ -5 \\ 2u \end{pmatrix} = 0 \Rightarrow 4 - 5u + (u-1)(2u) = 0$
 $u_1 = \frac{1}{2}$; $u_2 = 4$

4.2. $\frac{1}{2}x = -\sqrt{5-x^2}$

$$\frac{1}{2}x^2 = 5 - x^2$$

$$x^2 = 4$$

$$x = \pm 2$$

Probe: $x = +2$ ✓
 $x = -2$

4.3. $f(x) = x^4 e^{2x}$

$$f'(x) = 4x^3 e^{2x} + x^4 \cdot e^{2x} \cdot 2 = 0$$

$$\underbrace{2x^3}_{x=0} \cdot \underbrace{(2+x)}_{x=-2} \cdot \underbrace{e^{2x}}_{\neq 0} = 0$$